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SPECIAL REPORT

KDIGO 2025 ADPKD guideline: a commentary on diagnosis and management of hepatopancreatic manifestations by the ERA Working Group Genes & Kidney

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ABSTRACT

The Kidney Disease: Improving Global Outcomes (KDIGO) 2025 Clinical Practice Guideline for the Evaluation, Management, and Treatment of Autosomal Dominant Polycystic Kidney Disease (ADPKD) is a long-awaited development and the first KDIGO guideline addressing a genetic kidney disease. The approval of tolvaptan as targeted therapy for ADPKD in 2015 marked a pivotal advancement and motivated a structured approach in research and clinical care. The systemic nature of ADPKD guidance for the management of extrarenal manifestations is of special importance for nephrologists and requires interdisciplinary collaboration. In this regard, cysts in other organs, particularly the liver, are the most common extrarenal finding. While most liver cysts are incidental findings on routine imaging, both cyst infections and severe polycystic liver disease accompanying ADPKD come with a high disease burden and challenges in management. In contrast, pancreatic cysts in ADPKD are usually completely asymptomatic, but this raises questions about the need for follow-up. This commentary provides an overview of the most important recommendations and practice points on hepatopancreatic cysts provided by the KDIGO guideline with the aim to contribute to implementation and highlight remaining gaps in guidance, clinical care and research.

Keywords: ADPKD, liver transplantation, PLD, polycystic liver disease, somatostatin analogue

INTRODUCTION

Autosomal dominant polycystic kidney disease (ADPKD) is the most common genetic kidney disease—based on genetic prevalence—affecting >0.5 million people in Europe [1]. It is also the most frequent genetic cause of kidney failure and contributes 5–10% of all cases requiring kidney replacement therapy worldwide. Consequently, guidance towards optimal care in ADPKD is urgently needed. The Kidney Disease: Improving Global Outcomes (KDIGO) 2025 Clinical Practice Guideline for the Evaluation, Management, and Treatment of Autosomal Dominant Polycystic Kidney Disease is a major step forward in this regard and, as such, serves as the first KDIGO guideline for a genetic kidney disease. This guideline, an impressive text comprising 204 pages and 899 references, provides extensive guidance on a wide range of topics, including nomenclature, diagnostic strategies, prognostication and treatment of both the disease itself and its complications.

Considering the importance of the KDIGO guideline to improve patient care in ADPKD, the ERA Working Group Genes & Kidney

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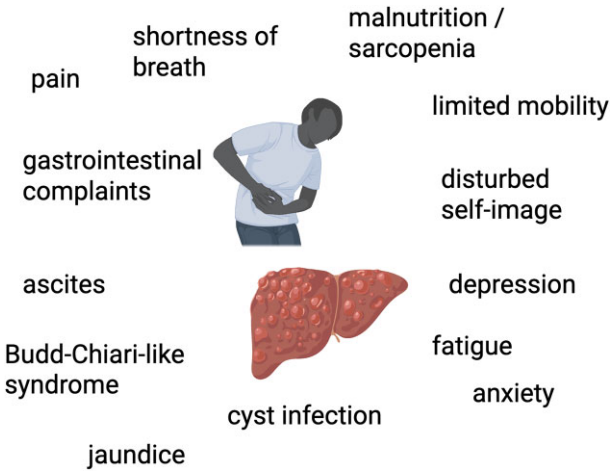

Figure 1: Symptoms and complications of PLD. PLD comes with various and severe symptoms and complications nephrologists should be aware of. Together with other patient characteristics and liver morphology, the set of symptoms guides patient management in PLD. Structured assessment of PLD-associated symptoms using validated tools such as the PLD-Q or POLCA questionnaires is a very important aspect in the care of patients with PLD. Created with biorender.com

and European Renal Best Practice (ERBP) will provide a series of commentaries addressing different topic areas to provide the European perspective and support implementation. Since extrarenal manifestations are an area requiring special attention and an interdisciplinary workup, the first commentary addresses the role of hepatopancreatic cysts in ADPKD. Extrarenal manifestations ranging from cysts of other organs to cardiovascular diseases and intracranial aneurysms lack targeted treatment opportunities to date. Nonetheless, or even more so, their management requires both guidance and interdisciplinary interaction. In this regard, cystic liver disease, as the most common extrarenal manifestation, requires special attention and is addressed by the guideline. More than 80% of patients with ADPKD show liver cysts on abdominal imaging [2], and a subset of 5–10% (primarily women) develops symptomatic polycystic liver disease (PLD). PLD may result in massive enlargement of the liver with ensuing symptoms, a significant impairment in quality of life and severe to life-threatening complications. Liver function usually remains normal in patients with PLD, even in the presence of extensive PLD. With the lack of approved treatment options, the severity of symptoms (Fig. 1) and potentially life-altering decisions such as liver transplantation, PLD is a special challenge in ADPKD and requires interdisciplinary approaches for optimal patient care. Pancreatic cysts are another known extrarenal manifestation of ADPKD and are nearly always asymptomatic. Central to management is to avoid anxiety (given the low risk of malignant degeneration) and a smart use of existing guidelines for adequate surveillance.

The purpose of this article, on behalf of the ERA Working Group Genes & Kidney and ERBP, is to provide an overview of the latest and most impactful aspects for the field, especially in the European context. This commentary serves to support implementation of the guideline, to highlight where action is needed in clinical care and to identify areas with an urgent need for research.

Diagnosis, screening and risk factors of PLD in ADPKD

A threshold of >10 cysts is often used to define PLD in clinical settings, although this remains an arbitrary criterion [3]. While PLD can occur without relevant renal involvement (ADPLD: pathogenic variants in *PRKCSH* or *SEC63*), this commentary focusses on PLD associated with ADPKD as discussed in the KDIGO guideline.

While liver cysts are detected in the majority of individuals with ADPKD—especially with increasing age—there is substantial variability in clinical expression [2]. Importantly, PLD severity does not correlate with kidney disease severity or genotype [4] and significant intrafamilial variability is observed.

Screening for liver involvement should be integrated into the routine workup of ADPKD (Box 1: Practice Points 5.1.1 and 5.1.3, Fig. 2 Care Pathway). MRI is the preferred imaging modality to assess both kidneys and liver. Imaging should—at least in patients with PLD or the first images obtained—cover the entire liver. Many patients, particularly women, are only referred to care once symptoms of hepatomegaly become disabling (Fig. 1), suggesting that earlier identification may improve management. Early screening is primarily valuable for identifying individuals at increased risk of progressive PLD, particularly young women with multiple liver cysts, in whom preventive strategies—such as avoiding oestrogen exposure—may be considered (Box 1: Practice Points 5.2.1.1, 8.1.2 and 8.1.4). The effect of sex hormones also comes with the clear implication that ovarian stimulation in the context of fertility treatments in patients with ADPKD should not be performed without prior liver imaging and counselling by an expert centre. In cases showing relevant PLD upon imaging, it is important to inform patients about the disease itself, potential acceleration of cyst growth in the context of pregnancies and symptoms that may occur and warrant interventions (Box 1: Practice Point 5.1.2). In contrast, individuals >40 years of age with only minimal hepatic involvement can generally be reassured, as progression to symptomatic PLD is unlikely at that stage.

Box 1: Diagnosis, screening and risk factors

Practice Point 5.1.1: When a computed tomography (CT) scan or magnetic resonance imaging (MRI) is performed for patients with ADPKD, liver images should be evaluated to characterize the severity of PLD.

Practice Point 5.1.3: People with ADPKD who are symptomatic due to possible hepatomegaly should have abdominal imaging performed to evaluate both liver and kidney volume.

G&K comment: Imaging in ADPKD should cover the entire liver, at least in the first MRI/CT obtained or in all instances in patients with PLD. Symptomatic patients with PLD will then need to be referred to expert centres. If MRI/CT is not readily available, ultrasound can be used to diagnose PLD. However, in symptomatic PLD, targeted treatment decisions will require MRI/CT.

Practice Point 5.1.2: When people with ADPKD are informed about the presence of liver cysts found on imaging, they should be advised of the likely outcomes and possible symptoms.

G&K comment: In cases with minimal liver involvement at an age >40 years, progression to symptomatic PLD is unlikely.

Practice Point 5.2.1.1: Women with ADPKD, particularly those with PLD, should be counselled about the benefits and potential harms of sex hormone therapy (see also Practice Points 8.1.2 and 8.1.4).

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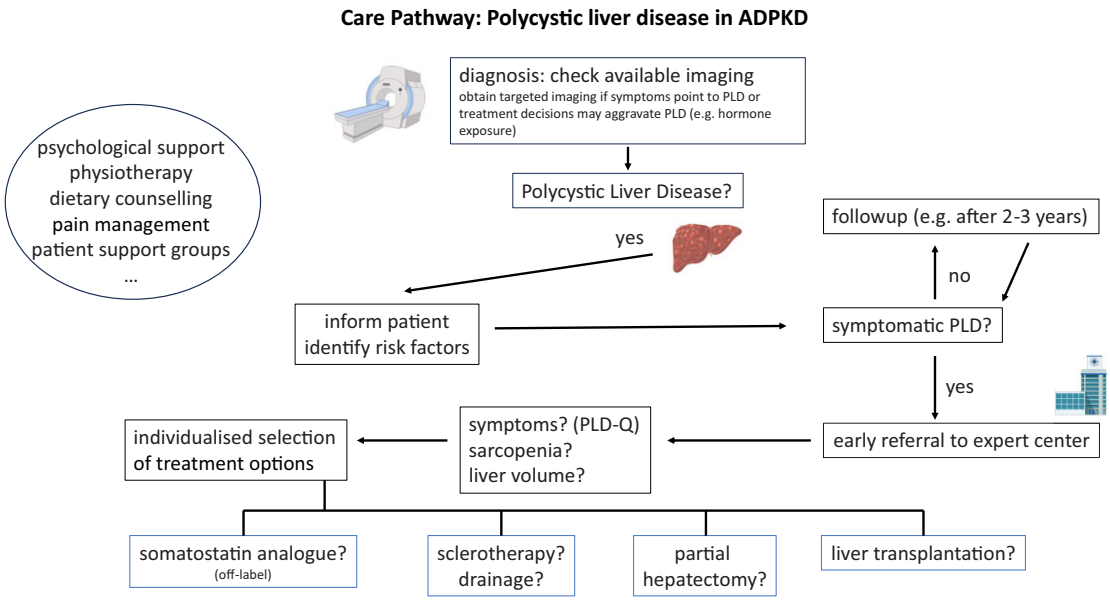

Figure 2: Pathway of care for patients with ADPKD-associated PLD. The liver should be examined in existing images of all ADPKD patients. Once PLD has been identified, the patient should be informed about this manifestation accompanied by obtaining information on potential PLD-associated symptoms. We suggest that patients with asymptomatic PLD should be invited for routine follow-up, e.g. after 2–3 years. All patients with symptomatic PLD should be referred to expert centres initiating structured assessment of symptoms, signs of malnutrition/sarcopenia and liver volume. These factors are the basis for individualized selection of treatment options in a shared decision-making process with the patient. In any case, further support involving psychologists, physiotherapy and dietary counselling may be needed. Created with biorender.com

G&K comment: To implement counselling awareness of disease is important, underlining Practice Point 5.1.1.

Diagnosis, screening and monitoring of pancreatic cysts in ADPKD

The 2025 KDIGO ADPKD guideline acknowledges pancreatic cystic lesions (PCLs) as an extrarenal manifestation but does not provide specific recommendations for their monitoring. These lesions may be simple cysts or premalignant neoplasms, such as intraductal papillary mucinous neoplasms (IPMNs). Although systematic screening for PCLs in ADPKD is not currently recommended, incidental detection of PCLs or IPMNs on imaging—often performed for Mayo classification or PLD assessment—may prompt further evaluation to rule out malignancy. In the absence of ADPKD-specific guidance, clinicians may resort to general PCL management guidelines [5–8]. These guidelines universally recommend surgical evaluation for PCLs exhibiting 'high-risk stigmata of malignancy' [7]. However, while recommendations diverge regarding the long-term monitoring of asymptomatic, low-risk PCLs (i.e. small simple cysts), all of them support follow-up over at least 5 years. Applying these guidelines in ADPKD comes with major challenges and a potentially excessive use of follow-up imaging considering the increased prevalence of PCLs (19–36%) in ADPKD compared with the general population [9, 10]. With increased use of state-of-the-art MRI, this prevalence will be increased even further. In ADPKD, pancreatic cysts are typically benign and rarely progress to malignancy. Routine imaging or intervention is not recommended unless cysts show high-risk stigmata. General guidelines for managing pancreatic cysts can be applied with caution, but less aggressive surveillance is suitable [10], and at careful discretion by the treating physician, to avoid unnecessary anxiety and unwarranted interventions.

Objective assessment of disease severity to characterize disease course and indications for therapeutic procedures

While many individuals with PLD remain asymptomatic, symptom burden becomes clinically relevant as liver cysts enlarge and exert pressure on adjacent structures (Fig. 1).

Specific treatment options are limited to the off-label use of somatostatin analogues as well as invasive measures, including liver transplantation, and carry substantial risk. Consequently, careful selection of patients, who may indeed benefit from these options, based on characterisation of both symptoms and liver morphology is important. On the level of liver imaging characteristics, several classification systems have been proposed to describe PLD severity. Gigot *et al.* [11] categorize disease based on cyst size and distribution to guide surgical decision-making. Other models integrate symptom severity, cyst characteristics and residual normal parenchyma [12] or propose a simple grading system based on cyst count alone [13]. More recently, classifications based on height-adjusted total liver volume (htTLV) have been proposed [14, 15]. For example, the HALT-PKD study (NCT00283686) defined mild, moderate and severe PLD using thresholds of <1000 ml/m, 1000–1800 ml/m and >1800 ml/m, respectively [14]. While liver volume correlates with symptom burden (with a cut-off of >1600 ml/m having been proposed to identify symptomatic patients [16]), this model does not integrate age or disease kinetics. To address this, a classification based on height-adjusted liver cyst volume (htLCV) and its annual growth rate was proposed defining five progression classes (A–E) corresponding to annual htLCV growth [15]. Sierks *et al.* [17] proposed a model that stratifies individuals into three progression groups based on age-adjusted htTLV: PG1 (<3.3% annual growth), PG2 (3.3–6.6%) and PG3 (>6.6%). These groups correspond with an increasing risk of PLD-related hospitalization [18]. While the tool is based on a large cohort study,

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external validation in other cohorts is needed to overcome limitations in generalizability.

Box 2: Objective assessment of disease severity to characterize disease course and indications for therapeutic procedures

Practice Point 5.1.4: Symptoms of PLD should be captured with the disease-specific symptom questionnaires Polycystic Liver Disease Questionnaire (PLD-Q) and Polycystic Liver Disease Complaint-specific Assessment (POLCA).

G&K comment: Since symptomatic patients should generally be referred to expert settings, we expect these tools to be used primarily in centres of expertise. Development of the PLD-Q focusses on ADPKD-related PLD and is provided in several languages. Symptoms should always be regarded in conjunction with liver volume and growth to make sure they are plausibly caused by PLD.

Practice Point 5.2.2.1: People should be advised that no specific diets are available to treat PLD and that they should follow the dietary recommendations and lifestyle advice for people with ADPKD and chronic kidney disease G1–G5.

Practice Point 5.2.3.1: Treatment for PLD should be performed in centres of expertise.

G&K comment: In our view, care for all patients with symptomatic PLD should be performed in centres of expertise, i.e. independent from the question of active treatment.

Practice Point 5.2.2.2: People with symptomatic PLD should be assessed for sarcopenia and malnutrition.

G&K comment: Sarcopenia is a late complication, i.e. signs towards malnutrition as well as any other PLD-associated symptoms should be identified before sarcopenia occurs to allow for early referral to an expert centre.

Practice Point 5.2.2.3: People with PLD and sarcopenia or malnutrition should be provided with intensive nutrition counselling and exercise rehabilitation.

Practice Point 7.1.5: Total kidney and liver weight derived from total kidney and liver volumes should be calculated and subtracted from the patient's total body weight for a more accurate assessment of weight and body mass index (BMI).

G&K comment: Correcting BMI is important both to avoid false diagnosis of obesity and missing malnutrition/sarcopenia.

Consequently, and based on the fact that the primary treatment target in PLD is symptom relief, assessment of symptom severity is key to guiding therapeutic decisions and should be performed using validated questionnaires (Box 2: Practice Point 5.1.4). This also includes measuring changes in symptoms over time. However, this is currently rarely implemented in clinical care. The guideline names two validated questionnaires for this purpose, the PLD-Q and the POLCA [19, 20]. Importantly, these tools outperform general quality-of-life instruments in capturing the nuanced impact of PLD. While both questionnaires address similar and overlapping items, the POLCA was primarily validated in ADPLD patients to assess the future need of liver transplantation while the studies establishing the PLD-Q included more patients with ADPKD-related PLD. To facilitate more widespread implementation of this tool in ADPKD, we provide the PLD-Q questionnaire as well as the analysis spreadsheet (https://doi.org/10.18716/MED2/PLD-Q). As for the PLD-Q, a score ≥32 identifies symptomatic patients eligible for treatment or inclusion in trials [19] and may thus also be considered a threshold to guide treatment selection (e.g. initiation of somatostatin analogues or transplantation). All symptomatic patients should be referred to expert centres for further evaluation of treatment options and counselling (see Patient Care Pathway in Fig. 2, Box 2: Practice Point 5.2.3.1), in line with the Clinical Practice Guidelines of the European Association for the Study of the Liver (EASL CPG, LoE 3, strong recommendation, 100% consensus) [21]. While we would like to motivate all physicians caring for patients with ADPKD to be aware of these questionnaires and implement their use whenever possible to quantify symptoms, implementation is likely to primarily occur in centres of expertise. Consequently, for the setting of the general nephrologist it is of utmost importance to generally identify symptomatic patients in a first step using simple questions. These questions should address key aspects of symptomatic PLD (Fig. 1) such as pain, loss of appetite, gastric fullness, nausea, dyspnoea, fatigue, limitations in sexual activity and anxiety.

As part of the evaluation of PLD-associated complications, objective assessment of sarcopenia and nutritional status is crucial, particularly in patients with advanced PLD or those being evaluated for liver transplantation (Practice Point 5.2.2.2). Several validated tools are available for this purpose [22]. The Skeletal Muscle Index (SMI), measured at the third lumbar vertebra on cross-sectional imaging, defines sarcopenia as an SMI <38.5 cm^2/m^2 in female patients and <52.4 cm^2/m^2 in male patients. Bioelectrical impedance analysis can also be used, with thresholds of <5.7 kg/m^2 in females and <7.0 kg/m^2 in males indicating sarcopenia. In our experience, longitudinal 24-h creatinine excretion and a drop to below the 10% percentile adjusted for age, sex and length can be useful to indicate developing sarcopenia. Grip strength (<18 kg for women, <26 kg for men) and mid-arm circumference (<23.1 cm for women, <23.8 cm for men) have also been proposed for detecting sarcopenia and malnutrition, and the latter is part of the Eurotransplant criteria for liver transplantation in PLD. Sarcopenia is a late complication of advanced PLD and should be avoided. Future research will have to clarify the role of changes in fat mass as an earlier indicator. While there are no PLD-specific dietary recommendations, dietary counselling in combination with physical exercise may limit progression to sarcopenia (Practice Point 5.2.2.3). Regarding malnutrition, it is important not to be misled by a normal BMI and to correct BMI for liver (and kidney) volume (Practice Point 7.1.5).

Non-invasive management and somatostatin analogue use

Once symptomatic PLD has been detected in a patient, early referral to an expert centre is warranted to allow for optimal patient counselling and determining treatment strategies (Box 2: Practice Point 5.2.3.1). A major risk factor that drives further liver growth is female sex hormones, specifically oestrogen, and patients should be counselled to avoid exogenous oestrogen [21] (Box 1: Practice Points 5.2.1.1, 8.1.2 and 8.1.4). Patients with PLD experiencing volume-related symptoms should be evaluated regarding potential interventional, surgical or medical treatment options in an individualized approach (Box 3: Practice Point 5.2.3.2). Serial MRIs can be helpful to characterize the natural disease course in an individual patient and to prove ongoing liver growth at the time of

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assessment (Box 3: Practice Point 5.2.3.2). However, limited reimbursement and availability of MRI, in general, and of liver volumetry specifically, may be a significant hurdle in this regard. Those with high liver volume due to multiple small, widespread liver cysts may be candidates for (off-label) therapy with somatostatin analogues.

Box 3: Management of PLD

Practice Point 5.2.3.2: People with ADPKD and PLD should receive treatment (i.e. medical and/or surgical, including minimally invasive treatments) if they experience cyst-related symptoms or complications that negatively impact their quality of life. Determination of treatment type should be based on symptoms, liver cyst characteristics, total liver volume and treatment availability.

G&K comment: Treatment decisions in PLD should be made by an expert centre. This holds for medical treatment with somatostatin analogues to reduce liver volume because of the side effects of this medication. The risk of surgical interventions should also not be underestimated. Portal hypertension and venous outflow obstruction are (relative) contraindications to partial hepatectomy. Treatment decisions should always consider whether symptoms observed are plausibly due to PLD. Serial imaging including liver volumetry may be of help to prove continued liver growth before exposing patients to the risk of therapy-related complications. This is of special importance in situations in which a cessation of liver growth or even a reduction of liver volume may occur, e.g. in postmenopausal women. Serial imaging with liver volumetry should therefore be considered at a low threshold in people in whom (it may be anticipated) medical or surgical treatment is or will be considered.

Practice Point 5.2.3.5: Ursodeoxycholic acid, mammalian target of rapamycin inhibitors and vasopressin-2 receptor antagonists should not be used to slow liver growth in people with PLD.

G&K comment: Ursodeoxycholic acid can be considered to prevent progression of de novo–formed gallstones in patients on somatostatin analogues.

Recommendation 5.2.3.1: We recommend prescribing long-acting somatostatin analogues (SSAs) in people with ADPKD and markedly enlarged polycystic livers with severe volume-related symptoms (1B).

Practice Point 5.2.3.4: When long-acting somatostatin analogues are prescribed, the effect on symptom burden and/or volume of polycystic livers and kidneys should be evaluated after 6–12 months.

G&K comment: In our expert opinion, a combination of repetitive PLD-Q assessment (e.g. at 6 and 12 months) and follow-up imaging at 12 months is a reasonable way to assess treatment response.

Practice Point 5.2.3.3: The administration of long-acting somatostatin analogues is usually well tolerated. Prescribing physicians should be aware of possible side effects (gastrointestinal symptoms, gallstones, hyperglycaemia, bradycardia).

G&K comment: All patients should be informed about relevant side effects before treatment. This includes awareness of diarrhoea, which is expected in most patients, often transient and can be mitigated by pancreatic enzyme substitution.

The strong recommendation (graded 1B) for long-acting SSAs in the current KDIGO guideline (Box 3: Recommendation 5.2.3.1) is based on randomized controlled trials (RCTs) that have shown SSA therapy can achieve a 2–8% reduction in total liver volume over 6–36 months [23–26]. However, there is considerable interindividual variability in liver volume response [26]. Younger premenopausal women are believed to benefit the most from SSA treatment, but robust predictors of treatment response are absent [27]. Furthermore, standardized recommendations on key aspects such as treatment initiation, monitoring, treatment duration and discontinuation are still lacking and remain the focus of ongoing initiatives within the European Reference Network on Rare Hepatological Diseases. Nevertheless, the current KDIGO guideline recommends an assessment of treatment response at 6–12 months (Practice Point 5.2.3.4), which is in line with the treatment duration in some of the RCTs. Although the KDIGO recommendation to discontinue SSA therapy in the absence of a beneficial effect seems reasonable, a stabilization of the symptom burden during disease phases of liver volume growth [18] may represent a minimal treatment response that is worth continuing.

As described in Practice Point 5.2.3.3 (Box 3), SSAs are usually well tolerated. However, some centres prescribe pancreatic enzymes as supporting medication at the start of therapy to mitigate the transient occurrence of diarrhoea due to (temporary) exocrine pancreatic insufficiency that some patients experience. Furthermore, potentially severe biliary side effects need to be discussed with patients before treatment initiation. In the DIPAK study (NCT01616927), 15% of patients on lanreotide developed new-onset gallstones, and administration of ursodeoxycholic acid, while not indicated to treat PLD per se (Box 3: Practice Point 5.2.3.5), has been suggested as a possibility in patients with incident gallstones upon SSA treatment [28]. One of the main limitations in clinical practice is the inconsistent access to SSA treatment for PLD patients across Europe. While SSAs also reduced polycystic kidney volume growth, they do not slow the decline in estimated glomerular filtration rate (eGFR) [29–31].

Liver cyst infections—diagnosis and management

The KDIGO guideline puts a strong emphasis on the clinical importance of liver cyst infections. This is very helpful in raising awareness for this common infectious complication that is likely to be underdiagnosed and easily mistaken. Importantly, an international Delphi consensus on the clinical management of liver cyst infections led by European experts has recently been published [32]. Both documents favour the use of [^{18}F]fluorodeoxyglucose positron emission tomography–computed tomography (PET-CT) as the imaging modality of choice to bolster the diagnosis of liver cyst infection (Box 4: Practice Points 5.3.1.1 and 5.3.1.2). It is important to note that isolated pain without fever and without relevant leucocyte and C-reactive protein (CRP) levels is not an indicator of cyst infection but may rather point to cyst rupture or haemorrhage (Box 4: Practice Point 5.3.1.3). Although largely overlapping with the KDIGO guideline (e.g. choice of first-line antibiotics, Box 4: Practice Points 5.3.2.1 and 5.3.2.2), there are subtle differences in terms of the time point of when to determine clinical response after the start of empiric antibiotic treatment: 48 h (KDIGO) versus 72 h (Delphi consensus). Both emphasize the need for sufficient

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treatment duration (KDIGO 4–6 weeks, Delphi consensus 6 weeks; Box 4: Practice Point 5.3.2.3), in line with the corresponding EASL CPG (4–6 weeks, LoE 4, strong recommendation, 100% consensus) [21]. Percutaneous cyst drainage is a reasonable option in specific risk settings and in case of clinical unresponsiveness to antibiotic treatment (Box 4: Practice Points 5.3.2.4 and 5.2.3.5). However, the KDIGO threshold to perform cyst drainage comes with a couple of questions towards implementation, e.g. when including immunocompromised patients per se, independent of the clinical context, as well as immediate drainage of infected cysts with a diameter >8 cm, a practice point that did not reach consensus in the Delphi framework. The latter also provides additional guidance beyond the KDIGO guideline when it comes to more detailed scenarios (e.g. non-accessible cysts, multifocal cyst infections, concomitant cholangitis).

Box 4: Liver cyst infections—diagnosis and management

Practice Point 5.3.1.1: Diagnosis of liver cyst infections should utilize culture data, advanced imaging and clinical signs and symptoms.

G&K comment: Fig. 33 of the guideline provides a helpful algorithm for the diagnostic strategy.

Practice Point 5.3.1.2: Imaging studies should be performed to determine the severity and location of a liver cyst infection.

G&K comment: Both diagnosis of cyst infections and determination of location are often difficult. PET-CT is the most sensitive approach and a helpful addition to the diagnostic strategy but can be false negative.

Practice Point 5.3.1.3: Empirical antibiotics should not be used to treat people with localized liver pain without fever who have normal white blood cell counts and CRP levels. Other causes such as cyst haemorrhage should be considered.

Practice Point 5.3.2.1: Empirical antibiotic treatment of liver cyst infections should target Gram-negative bacteria in the Enterobacteriaceae family.

Practice Point 5.3.2.2: Empirical antibiotic treatment of liver cyst infections should be initiated with a third-generation intravenous (i.v.) cephalosporin with or without a fluoroquinolone. After clinical stabilization, i.v. therapy can be switched to an oral fluoroquinolone, with adjustment according to culture results when available.

Practice Point 5.3.2.3: The duration of antibiotic therapy should be ≥4 weeks for liver cyst infection. Longer treatment periods may be required based on the response to therapy.

G&K comment: In line with other guidelines, our experience shows that many patients require a treatment duration of 6 weeks. Early reduction of CRP and symptoms may indicate cases in which 4 weeks is sufficient.

Practice Point 5.3.2.4: Percutaneous drainage of infected liver cysts <48 h after initiation of antibiotics may be reasonable in the presence of the following: isolation of pathogens that are unresponsive to antibiotic therapy from a cyst aspirate, immunocompromise in the patient, large infected hepatic cysts (>8 cm) or haemodynamic instability and/or signs of sepsis.

G&K comment: This recommendation suggests early drainage based on size (>8 cm) and immunosuppression status, but these factors alone did not reach consensus in the Delphi framework [31]. In particular, immunosuppression per se was not considered sufficient to warrant early drainage unless associated with a lack of clinical improvement. The Delphi panel favoured drainage primarily in non-responsive patients after 72 h of adequate antibiotics, emphasizing clinical evolution rather than static patient characteristics.

Practice Point 5.3.2.5: Infected liver cysts that do not respond to 48–72 h of antibiotic treatment should be evaluated further. Placement of a percutaneous drain should be considered for failure to improve, worsening symptoms or the presence of the risk factors listed, and the drain should be kept in place until drainage stops. In the case of deep cysts for which percutaneous drainage is not feasible, surgical drainage may be necessary.

Invasive measures and liver transplantation

The KDIGO guideline recommends directed therapy in case cyst-related symptoms impact quality of life significantly (Box 3: Practice Point 5.2.3.2). Surgical and interventional options have the appeal of a quick and finite treatment and come up in many conversations with patients who suffer from PLD volume-related symptoms [21]. PLD-related symptoms drive the decision to perform therapy, but choice may be guided by the growth of PLD over time as documented by volumetry, as well as cyst distribution. Stratification by gender is useful, as menopause in female patients with PLD is an inflection point of slower or even reversed liver growth. While surgical procedures have their merits, in the European space, these procedures are concentrated in only a few selected centres [33] (Box 3: Practice Point 5.2.3.2). Indeed, with PLD being a rare disease, only a few surgical teams have seen and treated more than a handful of patients. Massive PLD completely changes the architecture of the liver and in case of ADPKD this is compounded by the presence of polycystic kidneys. Consequently, volume-reducing surgery is a last-resort procedure, has to be weighed against liver transplantation and is usually reserved for selected patients with massive PLD who suffer from debilitating symptoms that result in a poor quality of life [34].

Aspiration sclerotherapy and/or fenestration of individual cysts

In symptomatic patients with PLD and large, uncomplicated liver cysts (typically ≥5 cm in diameter), imaging-guided interventional drainage followed by sclerotherapy can be considered (Box 3: Practice Point 5.2.3.2). Commonly used off-label sclerosing agents include polidocanol, sodium tetradecyl sulphate and ethanol [35, 36]. The technical success rates in terms of volume reduction and symptom relief are high and independent of the sclerosing agent used (72–100%), but the full therapeutic effect typically takes 3–6 months to achieve. Compared with surgical approaches, aspiration sclerotherapy is associated with lower morbidity and a shorter hospital stay. Once ascites develops, this procedure is usually no longer feasible. Some cysts may recur after sclerotherapy or—in case of very large volume—be difficult to access with this treatment modality. In these cases, targeted surgical fenestration may be an alternative. In this regard, some centres perform drainage to verify a relevant reduction in symptoms before moving to surgery.

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Partial hepatectomy (with cyst fenestration)

The KDIGO guideline mentions partial hepatectomy combined in a single procedure with fenestration of remnant liver cysts as a potential option in selected patients (Box 3: Practice Point 5.2.3.2), which should be restricted to expert centres (Box 2: Practice Point 5.2.3.1). In ADPKD it may be combined with nephrectomy to offer additional cyst volume reduction [37]. Recurrent cyst infections are a relative contraindication. The presence of at least one liver area with patent hepatic vasculature that is (relatively) spared from cysts is essential and morphology usually matches with Mayo Clinic PLD Classification Type C (liver volume >4500 ml) [38, 39]. The remaining liver segments should be able to meet with the postoperative metabolic demands. There is a general reluctance to advise partial hepatectomy combined with cyst fenestration due to the complexity. The widespread assumption that cyst proliferation will progress inevitably after surgery, abrogating its benefits, has been countered by a single-centre study that demonstrated sustained long-term benefit after partial hepatectomy (154 patients, 141 with ADPKD) as well as case reports using detailed CT liver volumetry [39, 40]. While there can be a significant improvement in Eastern Cooperative Oncology Group performance status as well as quality of life [38, 41], partial hepatectomy may also create complicated conditions for a potential subsequent transplantation (e.g. due to scarring and adhesions). Also, biliary leakage, persistent pleural effusion and/or ascites contribute to a significant morbidity rate (>40% Clavien ≥2) after partial hepatectomy. The reported perioperative mortality is 2.7–13.8% [38].

Box 5: Liver transplantation

Practice Point 5.2.3.6: People with PLD should be referred for liver transplantation in the event of massive PLD in the absence of contraindications or alternative treatment options.

G&K comment: Liver transplantation is a last-resort option in PLD associated with ADPKD. However, this option should be considered early enough (including referral to a transplant centre) to avoid major complications, including sarcopenia, and the priority on the liver transplant waiting list increases over time (depending on national criteria).

Practice Point 5.2.3.7: People with PLD should be referred for combined kidney–liver transplantation when an indication for liver transplantation is present and the person has severely impaired kidney function (eGFR <30 ml/min/1.73 m^2).

G&K comment: Currently, access to both liver transplantation and combined kidney–liver transplantation in Europe is complicated by the lack of harmonization of both centre-specific strategies and national criteria for listing. Reaching better harmonization in this area is an important action point for the future and will also enable improved guidance for the field. It is important to keep in mind that eGFR may overestimate kidney function in malnourished patients and that direct GFR measurement should be considered in individuals with an eGFR <45 ml/min/1.73 m^2.

Liver transplantation

The KDIGO guideline (Box 5: Practice Point 5.2.3.6) indicates that liver transplantation should be considered for patients with massive PLD once alternative options have been exhausted. Liver transplantation in ADPKD serves to improve the quality of life and aims to avoid or manage major complications of massive PLD [34]. The reMELD-Na scoring system, used to prioritize patients for liver transplantation, has little value in PLD, as most parameters contributing to reMELD-Na (sodium, international normalized ratio, bilirubin, creatinine) remain normal even in advanced cases. The Organ Procurement and Transplantation Network (USA) allows PLD patients with severe symptoms, not suitable for resection/fenestration, to receive MELD exception points [42]. In Europe, there is considerable variation in liver transplantation strategies in patients with PLD, and uniform PLD-specific criteria for isolated and combined liver–kidney transplantations are absent. The Eurotransplant Liver Allocation Manual lists ascites, variceal bleeding, Budd–Chiari-like syndrome and severe malnutrition among the exceptional MELD criteria for PLD [43]. However, the applicability of these criteria differs by member country and may be entirely different in non-Eurotransplant countries. As an example, in France, the allocation of points depends on the lack of alternative options, requirement of kidney replacement therapy, abdominal wall complications and cyst infection with multidrug-resistant bacteria.

While the outcomes of liver transplantation for PLD are excellent, it is worth acknowledging that the procedure itself carries significant perioperative risks. Hepatectomy in massively enlarged polycystic livers is technically challenging and is typically performed only in a limited number of highly experienced transplant centres. The European Liver Transplant Registry reported good patient survival at 1 (90%) and 10 (78%) years without appreciable graft loss [73% (non–death censored) graft survival at 10 years). Importantly, in patients with an eGFR <30 ml/min/1.73 m^2, combined liver–kidney transplantation should be considered (Box 5: Practice Point 5.2.3.7). Progressive decline in kidney function is expected as part of the natural history of the disease, and liver transplantation—particularly in the setting of perioperative stress and calcineurin inhibitor exposure—may further accelerate renal deterioration. Currently there is no consensus in Europe whether simultaneous liver–kidney transplantation or a sequential liver-first approach is preferred, highlighting that structured analyses of the outcomes associated with each strategy would be an important asset.

In any case, timely referral of patients with severe PLD to transplant centres (for a first contact) and expert opinion remains crucial. Patients are often referred only once all other options have been exhausted; however, delaying referral until the onset of profound functional decline—characterized by severe malnutrition, sarcopenia and marked reduction in daily activity—may compromise transplant eligibility and outcome.

CONCLUSION

PLD is among the most common and clinically most relevant extrarenal manifestations of ADPKD. While severe PLD is rare, it comes with significant symptoms for the patients affected and limited treatment options and should be assessed by expert centres as early as possible. The combination of kidney and liver enlargement adds to the disease burden. Consequently, awareness of PLD among nephrologists and assessment of symptoms using validated tools is crucial to optimize care for these patients in an interdisciplinary setting with hepatologists, (transplant) surgeons and additional disciplines depending on the individual complications. The KDIGO ADPKD guideline provides crucial guidance regarding PLD in ADPKD and, together with existing

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recommendations from the field of hepatology, will help to improve patient care. While standards of care may always show national differences, especially in a multinational setting like the European Union, evidence-based harmonization remains an important goal as a basis for improved patient care and research. Importantly, future research is urgently required to develop new therapeutic options for PLD.

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